

Freeze-Dried Mango

Processing Technology, Benefits & Export Potential

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Key Facts at a Glance

- Freeze-drying removes water via sublimation — ice converts directly to vapor below the triple point (~6.1 mbar) without passing through a liquid phase.
- The process runs in three stages: rapid freezing (~-40°C), primary drying (sublimation, 10–30+ hours), and secondary drying (desorption of bound moisture).
- Properly processed freeze-dried mango achieves water activity below 0.3 and final moisture content of 1–4%, giving it a shelf life of 18–24+ months when properly packaged.
- Chaunsa and Sindhri are the most widely used Pakistani cultivars for freeze-drying, due to low fiber content, high sweetness, and firm slicing texture.
- Only an estimated 6–8% of Pakistan's total mango production is currently exported — freeze-drying offers a route to capture more value from the remainder.

The 12-Step Processing Flow

1. Raw Material Reception	2. Sorting & Grading
3. Washing & Sanitation	4. Peeling
5. Slicing / Cubing	6. Pretreatment (optional)
7. Rapid Freezing (~-40°C)	8. Primary Drying (Sublimation)
9. Secondary Drying (Desorption)	10. Moisture Testing & QC
11. Vacuum Pack / N ₂ Flush	12. Storage & Quality Assurance

Freeze-Drying vs. Other Drying Methods

Characteristic	Freeze-Drying	Hot-Air Drying	Vacuum Drying
Operating temp.	Below 0°C (primary)	55–90°C	30–60°C
Vitamin C retention	80–90%+	40–60%	55–70%
Color retention	Excellent	Poor–fair	Fair
Final moisture	1–4%	10–20%	5–15%
Shelf life	18–24+ months	6–12 months	9–15 months

Pakistani Mango Varieties for Freeze-Drying

Variety	Region	Fiber	Freeze-Drying Suitability
Sindhri	Sindh	Low (fiberless)	Excellent
Chaunsa	Punjab	Very low	Excellent
Anwar Ratol	Punjab	Low	Very good (premium/aroma)
Langra	Punjab	Moderate	Good
Dusehri	Punjab	Low-moderate	Good (powder/puree)
Saroli	Sindh	Low	Very good

This is a condensed summary. For the complete article — including the full industrial process explanation, sublimation science, industrial applications, packaging and export standards, Pakistan's mango industry analysis, and 15 frequently asked questions — visit the full article at muhammadhassanakram.me/articles/freeze-dried-mango-processing.html